

TP1 Input Validation

Team Project 1 — Foundations Application

This document identifies every text input field in the TP1 application and the validation rule(s) enforced on that field. Every field listed below is checked for excessive input length before any other processing takes place, which protects the application against a classic attack pattern: submitting an unreasonably long string to probe for a crash-based vulnerability.

1. Username

Enforced by `userNameRecognizerTestbed.UserNameRecognizer.checkForValidUserName()`, a finite-state-machine validator. Used on: New Account creation, First Admin setup, and Update Username.

Rule	Requirement	Error Message
Starting character	Must be a letter (A–Z or a–z)	"A UserName must start with A-Z, a-z."
Body characters	Letters, digits, or the separators . & - _	"A UserName character may only contain the characters A-Z, a-z, 0-9."
Separator rule	A separator must be followed by a letter or digit (no doubled or trailing separators)	"A UserName character after a separator character must be A-Z, a-z, 0-9."
Minimum length	At least 4 characters	"A UserName must have at least 4 characters."
Maximum length	No more than 16 characters	"A UserName must have no more than 16 characters."
Empty input	Rejected immediately	"The input is empty"

2. Password

Enforced by `passwordEvaluator.PasswordEvaluator.evaluatePassword()`, adapted from the PasswordEvaluationTestbed-F26 Model class, with a maximum-length rule added. Used on: New Account creation, First Admin setup, and Update Password.

Rule	Requirement	Notes
Minimum length	At least 8 characters	Matches the original testbed requirement
Maximum length	No more than 32 characters	Added for TP1 — rejects the password before any character is examined, preventing a crash-probing attack with an extremely long string

Rule	Requirement	Notes
Upper case	At least one A–Z character	
Lower case	At least one a–z character	
Numeric digit	At least one 0–9 character	
Special character	At least one of ~ ` ! @ # \$ % ^ & * () _ - + = { } \ ; ' " < > , . ? /	
Passwords match	The two password entry fields must be identical	Checked separately by the account creation/setup controller

3. Name and Email Address Fields

First Name, Middle Name, Last Name, Preferred First Name, and Email Address (all on the Update Account page) are free-form fields with no required character format. Each is checked against a shared maximum length constant (**MAX_TEXT_FIELD_LENGTH = 50**) before the update is applied to the database.

Field	Rule	Error Message
First Name	Must not exceed 50 characters	"First Name must not exceed 50 characters."
Middle Name	Must not exceed 50 characters	"Middle Name must not exceed 50 characters."
Last Name	Must not exceed 50 characters	"Last Name must not exceed 50 characters."
Preferred First Name	Must not exceed 50 characters	"Preferred First Name must not exceed 50 characters."
Email Address	Must not exceed 50 characters	"Email Address must not exceed 50 characters."

Why only a length check for these fields? Unlike Username and Password, these fields are not security-critical and do not gate access to the system — a name or email typo does not create a vulnerability the way a malformed username or a weak password would. The length limit alone is sufficient to block the crash-probing attack the assignment describes, without imposing character-format restrictions (e.g. hyphenated names, accented characters) that would unnecessarily burden legitimate users.

4. Invitation Code

Entered on the login page to begin New Account setup. Validated indirectly: the code is looked up in the database, and an invalid/unrecognized code triggers an "Invalid Invitation Code" alert, so a code that does not exist can never be used to create an account.

Known limitation: this field does not currently have an explicit maximum-length check before the database lookup occurs. Because the lookup uses a parameterized query, an overly long input cannot corrupt the query,

but adding an explicit length guard here would bring this field in line with the length-check standard applied to every other field, and is planned as a follow-up improvement.